

Elevation of boiling point of the solvent

$$1. \quad (a) K_b = \frac{M_1 RT_0^2}{1000 \Delta H_V} = \frac{18 \times 1.987 \times (373)^2}{1000 \times 9700} = 0.513^\circ C$$

$$2. \quad (b) \Delta T_b = imk_b = 0.52 \times 1 \times 2 = 1.04. \\ \therefore T_b = 100 + 1.04 = 101.04^\circ C.$$

$$3. \quad (c) K_b = \frac{\Delta T_b}{m} = \frac{0.1 \times 100}{\frac{1.8}{180} \times 1000} = 1 K/m.$$

$$4. \quad (d) m = \frac{K_b \times w \times 1000}{\Delta T_b \times W} = \frac{2.16 \times 0.15 \times 1000}{0.216 \times 15} = 100.$$

5. (b) Due to higher pressure inside the boiling point elevated.

6. (b) Dissolution of a non-volatile solute raises the boiling pt. of a liquid.

7. (b) As we know that

$$\text{Boiling point} \propto \frac{1}{\text{vapour pressure of liquid}}$$

Hence, on decreasing vapour pressure, boiling point will increase.

$$8. \quad (b) \Delta T_b = \frac{100 \times K_b \times w}{m \times W} \quad \therefore 0.52 = \frac{100 \times 5.2 \times 6}{m \times 100}$$

$$m = \frac{100 \times 5.2 \times 6}{0.52 \times 100} = 60.$$

9 (b) $T_1 - T_2$

Explanation:

Elevation in boiling point is:

$$\Delta T_b = T_1 - T_2$$

where:

- T_1 = boiling point of solution
- T_2 = boiling point of pure solvent.



10. (b) Elevation in a boiling point is a colligative property as it depends upon the number of particles.

$$\Delta T_b \propto n$$

For sucrose, $n = 1$, $\Delta T_b = 0.1^\circ\text{C}$

For NaCl , $n = 2$, $\Delta T_b = 0.2^\circ\text{C}$

11. (b) $\Delta T_b = K_b \times m$ or $K_b = \Delta T_b / m$

12. (c) $\Delta T_b = K_b \times m = 0.513 \times \left(\frac{0.1}{200} \times 1000\right)$
 $= 0.2565^\circ\text{C}$, $T_b = 100.256^\circ\text{C}$

- 13 (c) $8.3 \text{ J mol}^{-1} \text{ K}^{-1}$

Explanation:

The SI value of gas constant is:

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

- 14 (b) **Boiling point**

Explanation:

A liquid boils when its vapour pressure equals external atmospheric pressure.

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Given:

- Mass of $\text{CuCl}_2 = 13.44 \text{ g}$
- Molar mass = 134.4
- $K_b = 0.52 \text{ K molal}^{-1}$

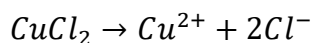
Moles of CuCl_2 :

$$\frac{13.44}{134.4} = 0.1 \text{ mol}$$

Molality:

$$m = 0.1$$

Since:



Total ions produced = 3

Assuming complete ionization:

$$i = 3$$

Now,

$$\begin{aligned}\Delta T_b &= iK_b m \\ \Delta T_b &= 3 \times 0.52 \times 0.1 \\ &= 0.156 \approx 0.16\end{aligned}$$

(a) 0.16

16. (d) $\Delta T_b = \frac{K_b \times w \times 1000}{m \times W}$

$$m = \frac{K_b \times w \times 1000}{\Delta T_b \times W} = \frac{2.53 \times 10 \times 1000}{1 \times 100} = 253g.$$

17. **Explanation:**

Elevation in boiling point depends on number of moles of solute.

For urea:

$$\text{moles of urea} = \frac{1}{60}$$

For glucose:

$$\text{moles of glucose} = \frac{3}{180} = \frac{1}{60}$$

Both solutions contain equal number of moles in same volume of solvent.

Therefore, elevation in boiling point is same.

$$100^\circ\text{C} + 0.25^\circ\text{C} = 100.25^\circ\text{C}$$

(c) 100.25°C

18. (b) Common salt is non-volatile and rises the b.pt.



19. (b) In the process of evaporation, high energy molecules leave the surface of liquid, hence average kinetic energy and consequently the temperature of liquid falls.
20. (a) The boiling occurs at lower temperature if atmospheric pressure is lower than 76 cm Hg.

21 (b) **Molal elevation constant**

Explanation:

The boiling point elevation constant K_b is defined as the elevation in boiling point produced when 1 mole of solute is dissolved in 1 kg of solvent.

$$\Delta T_b = K_b m$$

For $m = 1$,

$$\Delta T_b = K_b$$

Hence K_b is called molal elevation constant.

22 (d) **Chloroform**

Explanation:

For 1 molal solution:

$$\Delta T_b = K_b$$

So maximum boiling point elevation occurs for the solvent having highest K_b value. Chloroform has the highest molal elevation constant among the given solvents.

23. (b) $BaCl_2$ furnishes more ions than KCl and thus shows higher boiling point $T_1 > T_2$.

